COS 350 Lecture 1

Introduction to:

- Systems Programming
- C
- Unix

Application Programs	COS 160, 161, 285, …
Software Tools & Libraries	COS 350 Systems Programming
Operating System Kernal	COS 450 Operating Systems
Hardware	COS 250 & 255 Computer Organization

C

Developed 1972
by Dennis Ritchie at AT&T

Goals:

- simple
- very efficient
- Intended use: Operating Systems Implementation

Design:

- operations & types match HW
- direct memory addressing
- no bounds checking

Modern Descendants:
C++, Java, C#

Unix

Developed early 1970's
by Ken Thompson at AT&T

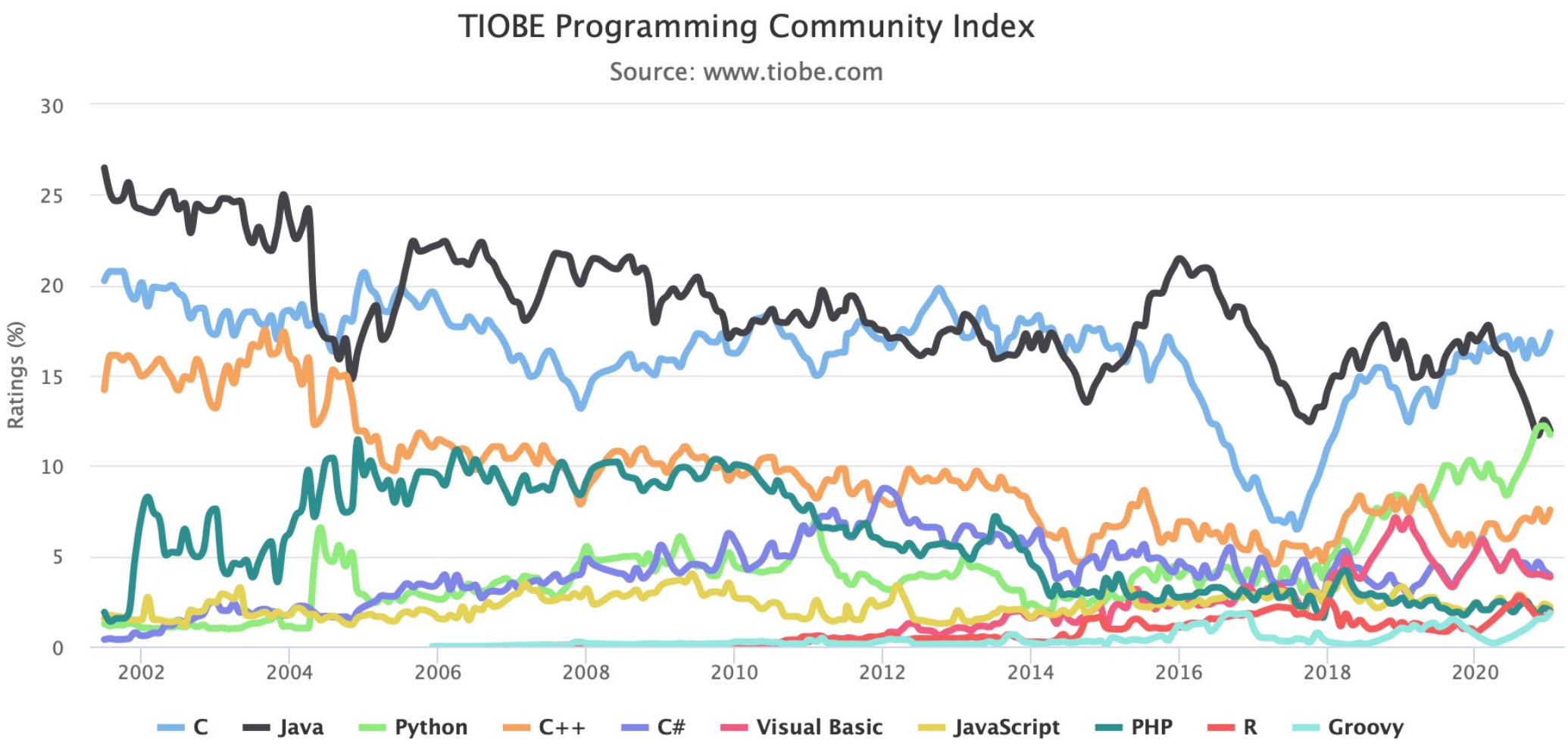
Goals:

- multi-user
- multi-tasking
- virtual memory
- portable

Design:

- small single task programs
- programs communicate
- written in C

Modern Descendants:
linux, Mac OS X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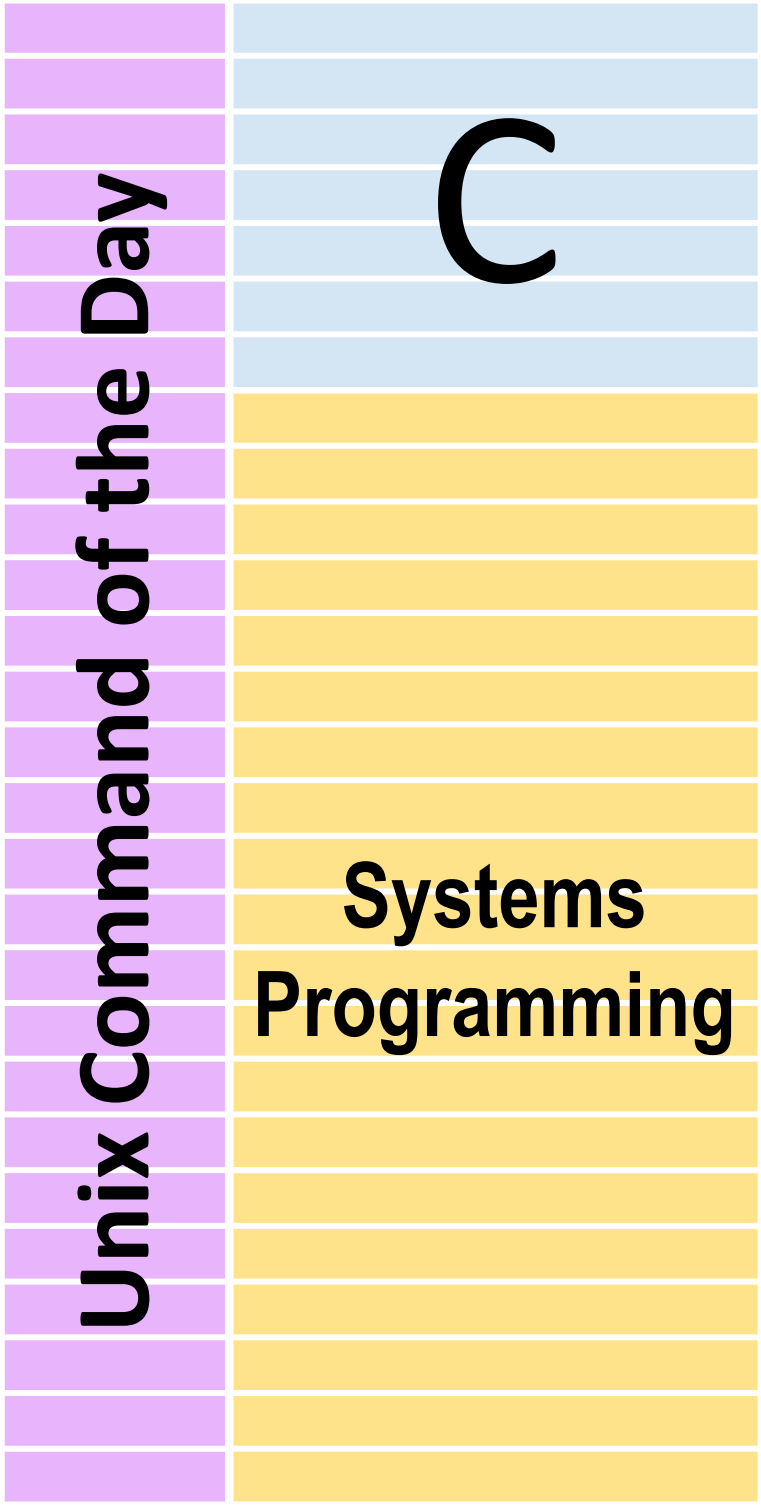
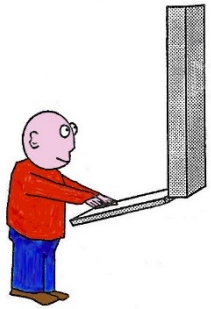
Course Structure





hello.c

	Basic Types	Size	Signed Range	**unsigned** Range
integer types	**char**	8 bits	-128 to 127	0 to 255
	short	16 bit	-32768 to 32767	0 to 65535
	int	16 or **32** bit	±2 billion	0 to 4 billion
	long	32 or **64** bit	$\pm 9 \times 10^{18}$	0 to 18×10^{18}
	long long	64 bit	$\pm 9 \times 10^{18}$	0 to 18×10^{18}
			Range	Precision
floating point	**float**	32 bit	$\pm 10^{\pm 38}$	6 digits
	double	64 bit	$\pm 10^{\pm 308}$	15 digits

What is missing?

C has no boolean type,

but any integral type can be used as a boolean.

0 is false

anything else is true

```
int isMinor = age < 21;
if (isMinor)
   printf("Sorry, you aren't old enough.");
```

Many C programmers use this to simplify expressions:

```
while (count != 0) …
```

can be shortend to:

```
while (count) …
```

	C		Java	
	value types	pointer types	value types	reference types
primative types	char c;	char *c;	char c;	
	int i;	int *i;	int i;	
	double d;	double *d;	double d;	
array types	int a[10];	int *a;		int[] a;
	char buf[80];	char *string;		char[] buf;
aggregate types	struct timeval tv;	struct timeval *tv;		String s;
				Point p;
functions	int func() { … }	int (*func)();	int func() { … }	

```c
#include <stdio.h>

int main() {
  int age;
  printf("What is your age? ");
  scanf("%d", &age);
  int isMinor = age < 21;
  if (isMinor)
    printf("Sorry, you aren’t old enough.\n");
}
```

```
unix> a.out
What is your age? 19
Sorry, you aren’t old enough.
```

```c
#include <stdio.h>

int main() {
  int month, day, year;
  printf("Enter your birthday MM/DD/YYYY\n");
  int result = scanf("%d/%d/%d", &month, &day, &year);
  if (result != 3)
    printf("Error on input\n");
  else
    printf("You entered %02d/%02d/%04d\n",
           month, day, year);
}
```

```
unix> a.out
Enter your birthday MM/DD/YYYY
5/14/1985
You entered 05/14/1985
```

Reasons to use pointers

1. Efficient passing of arguments
2. Passing arguments you want to modify
3. Multiple return values
4. Fast streamlined code
5. Linked data structures

Common UNIX Commands

ls	list files
cp src-file dest	copy file
mv src-file dest	move file
rm file	remove file
man cmd man –k keyword	display manual page
cat file	type file to screen
emacs file	text or program editor
gcc file.c	compile c program
a.out	execute default result
exit	exit shell

NAME

man - an interface to the on-line reference manuals

SYNOPSIS

man [section] page
man -k regexp

DESCRIPTION

man is the system's manual pager. Each page argument given to man is normally the name of a program, utility or function. The manual page associated with each of these arguments is then found and displayed. A section, if provided, will direct man to look only in that section of the manual.

EXAMPLES

man ls
Display the manual page for the item (program) ls.

OPTIONS

-k, --apropos
Equivalent to apropos. Search the short manual page descriptions for keywords and display any matches. See apropos(1) for details.

Section	Description
1	General commands
2	System calls
3	C library functions
4	Special files (usually devices, those found in /dev) and drivers
5	File formats and conventions
6	Games and screensavers
7	Miscellanea
8	System administration commands and daemons

UNIX Commands

mkdir dir	make a directory
cd dir	change working directory
rmdir dir	remove directory
lpr file	print file
a2ps file	print two pages side-by-side
ls -l	list files with “long” listing option
more file	display file one screenful at a time